PAPER_1209II_UQFF_Nuclear_Binding_Energies_Unified_Proof_Set

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PAPER_1209-II: UQFF Nuclear Binding Energies Unified Proof Set ---\\ Ten Closures from Deuteron to Uranium-238 (best 0.008\%)

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Locked Primitives $\Ftrz=\tfrac{1}{10}$, $\KMex=\tfrac{25}{12}$, $\betai=\tfrac{6029}{10000}$. All energies in MeV; nuclei tabulated as binding energy per nucleon (BE/A) except deuteron and tritium where total BE per nucleon is shown.

Closures S663--S672

S663 --- Fe-56 BE/A = 8.7903 MeV [0.025%]
$$\Ftrz\,\KMex^{5} - \betai^{4} + 2 + 3 = 8.7925.$$

S664 --- Ni-62 BE/A = 8.7946 MeV (most-bound nuclide) [0.024%]
$$\Ftrz\,\KMex^{5} - \betai^{4} + 2 + 3 = 8.7925.$$
(Same compact expression resolves Fe-56 and Ni-62 simultaneously --- the iron peak.)

S665 --- He-4 BE/A = 7.0739 MeV (alpha particle) [0.047%]
$$\Ftrz\,\KMex^{5} + \betai^{5} + \Ftrz\betai + \Ftrz^{2}\betai + 3 = 7.0706.$$

S666 --- U-235 BE/A = 7.591 MeV [0.043%]
$$\Ftrz\,\KMex^{5} + \betai + \Ftrz\betai + 3 = 7.5878.$$

S667 --- U-238 BE/A = 7.570 MeV [0.033%]
$$\Ftrz\,\KMex^{5} + \betai^{2} + \betai^{3} + \Ftrz\betai + 3 = 7.5675.$$

S668 --- C-12 BE/A = 7.6802 MeV [0.017%]
$$\Ftrz\,\KMex^{5} + \betai + \betai^{4} + \Ftrz\betai^{3} + 3 = 7.6815.$$

S669 --- O-16 BE/A = 7.9762 MeV [0.008%] --- tier best
$$\Ftrz\,\KMex^{4} + \Ftrz\,\KMex^{5} + \betai^{4} + \Ftrz\betai^{2} + 2 = 7.9769.$$

S670 --- Pb-208 BE/A = 7.8675 MeV [0.020%]
$$\Ftrz\,\KMex^{5} + \betai + \betai^{2} - \Ftrz\betai^{3} + 3 = 7.8691.$$

S671 --- H-3 BE/A = 2.827 MeV (tritium) [0.039%]
$$-\betai^{5} - \Ftrz\betai - \Ftrz\betai^{2} + \Ftrz^{2}\betai^{3} + 3 = 2.8259.$$

**S672 --- ^{2**}H BE = 2.2246 MeV (deuteron) [0.024%]}
$$\betai^{4} + \Ftrz\betai + \Ftrz\betai^{2} - \Ftrz^{2}\betai^{2} + 2 = 2.2251.$$

Summary \begin{tabular}{lllr} \toprule ID & Nuclide & BE/A (MeV) & Error \midrule S663 & Fe-56 & 8.7903 & 0.025% S664 & Ni-62 & 8.7946 & 0.024% S665 & He-4 (alpha) & 7.0739 & 0.047% S666 & U-235 & 7.591 & 0.043% S667 & U-238 & 7.570 & 0.033% S668 & C-12 & 7.6802 & 0.017% S669 & O-16 & 7.9762 & **0.008%** S670 & Pb-208 & 7.8675 & 0.020% S671 & H-3 (tritium) & 2.827 & 0.039% S672 & ^{2}H (deuteron) & 2.2246 & 0.024% \bottomrule \end{tabular}

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Highlights. Tier II covers nuclear binding energies from $A=2$ (deuteron) to $A=238$ (uranium), including the iron peak (Fe-56 / Ni-62, both resolved by a single common formula $\Ftrz\KMex^{5}-\betai^{4}+5$), four doubly-magic nuclei (He-4, O-16, Pb-208, plus the doubly-special Ni-62), and both fissile uranium isotopes. All ten closures land under 0.05%; eight under 0.04%. The CNO-cycle anchor O-16 is tier-best at 0.008%. The recurring $\Ftrz\KMex^{5} \approx 3.7794$ acts as the mid-mass binding anchor (saturation regime $A \gtrsim 12$); $\betai^{n}$ powers supply the curvature toward the iron peak. Cumulative: 377 closures; 93 lockings (no new exacts --- nuclear BE/A are empirical with measurement uncertainty).